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Experiment (1) Report

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Faculty of Engineering - Ain Shams University
Electronics and Electrical Communications Engineering Department

ECE352s Analog and Digital Communication Systems

Junior - Electronics and Electrical Communications

Experiment (1)

Introduction to Communication Systems Lab

Spring 2026

1. Objectives:

- Get familiar with cathode ray oscilloscope (CRO) equipment and visualizing real time signals.
- Get familiar with the Telecoms-Trainer 101 KIT.
- Visualize master signals from the KIT using CRO.

2. Equipment:

- Digital CRO
- Telecoms-Trainer 101 KIT

3. Part 1: Setting up an Oscilloscope

The cathode ray oscilloscope (CRO) or simply a scope is one of the most versatile pieces of test equipment in the electronics industry, it gives us a visual display of the amplitude (voltage), and period of inspected signal.

The experiment gives you a procedure that will setup most scopes ready to show a stable 2 KHz 4 V_{p-p} signal on the scope's screen every time.

Procedure:

Switch on the scope, a trace should appear on display.

1. General controls:
   a. Set the intensity control to about ¾ of its travel.
   b. Set the mode control to CH1 position.
   c. Adjust the focus control for sharp trace.
2. Vertical controls:
   a. Set the input coupling control for both channels (CH1, and CH2) to AC coupling.
   b. Set the vertical attenuation control for both channels to 1V/div.
   c. Set the vertical position control for both channels to about the middle of their travel.
3. Horizontal controls:
   a. Set the horizontal time base control to 0.5 mS/div.
   b. Set the horizontal position control to about the middle of its travel.
4. Triggering controls:
   a. Set the sweep mode control to the AUTO position.
   b. Set the trigger source control to CH1 position.
   c. Set the trigger source coupling control to the AC position.
   d. Set the slope (or sync) control to the "+" position.

5. Use the oscilloscope lead (probe) to connect the CH1 input to the scope's CAL output.
6. Use Channel 1's Vertical Attenuation control to make the waveform as big on the screen as possible without it going past the top and bottom lines.
7. Use the Horizontal Position control to align the top of the waveform with the center vertical line on the screen.
8. Use Channel 1's Vertical Position control to move the bottom of the waveform so that it touches any one of the horizontal lines on the screen.
9. Your display should now look something like Figure 1.
10. Count the number of divisions from the bottom of the waveform to the top. **Note**: The sub divisions are worth 0.2.

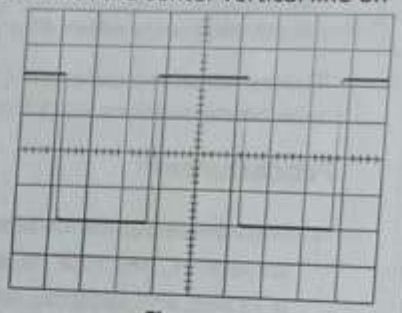


Figure 1

11. Multiply this number by the Vertical Attenuation control's setting. For example: If you counted 6.6 divisions and the Vertical Attenuation control's setting is 0.5V/div, then multiply 6.6 by 0.5V. Using these values, the peak-to-peak voltage is 3.3V.
12. Record your measurement: CAL output's peak to peak voltage = ...2.08 V
13. Use the Horizontal time base control to make the scope's CAL signal as wide on the screen as possible while still showing one complete cycle.
14. Set Channel 1's Input Coupling control to the GND position.
15. Use Channel 1's Vertical Position control to align the trace with the horizontal line across the middle of the screen.
16. Return Channel 1's Input Coupling control to the AC position.
17. Use the Horizontal Position control to align the start of the waveform with the first vertical line on the screen.
18. Your display should now look something like Figure 2.
19. Count the number of divisions for one complete cycle of the waveform, hence multiply this number by the horizontal time base control's setting. **Note**: The subdivisions are worth 0.2.

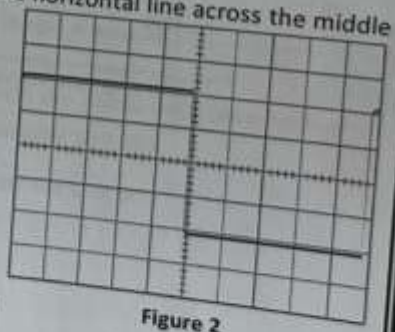


Figure 2

20. Record your measurement: CAL output's period = ...1ms..... , CAL output's frequency = ...1KHz

4. Part 2: Introduction to the Telecoms-Trainer 101

The Emona Telecoms-Trainer 101 is designed to help users learn about communications and telecommunications principles by bringing block diagrams to life, these blocks are simplified representations of complex circuits where each block represents a part of the circuit performing a specific task (e.g., adder, filter, phase shifter).

The trainer uses a collection of blocks called "modules" that can be combined to implement various telecommunications block diagrams.

Procedure:

1. The Master Signals module is an AC signal generator or oscillator, it has six outputs, an analog sine wave of 2K, and 100K Hz, and analog cosine wave of 100 KHz, and digital square wave of 2.083 KHz, 8.33 KHz, and 100 KHz.

   a. Connect the setup as shown in figure 3.
   b. Setup the scope trigger to auto mode, adjust the time base control to view 2 cycles of the masters' module 2 KHz sine wave output.
   c. Measure the amplitude (peak to peak), period and frequency of the output signal.
   d. Repeat steps a to c for the 100 KHz sine, and cosine waves in the master's module.

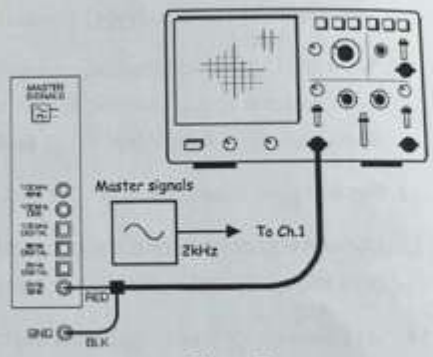


Figure 3

Master's Signal	Output voltage	Period	frequency
2 KHz sine wave	3.88	480µs	2.08KHz
100 KHz sine wave	3.84	10µs	100KHz
100 KHz cosine wave	3.96	10µs	100KHz

3. Several communications and telecommunications systems require that signals be added together. The Adder module has been designed for this purpose.

a. Connect the set-up shown in Figure 6

b. Set the Adder module's *G* control for (*Input A*) to about the middle of its travel, while turn its *g* control (for *Input B*) fully anti-clockwise.

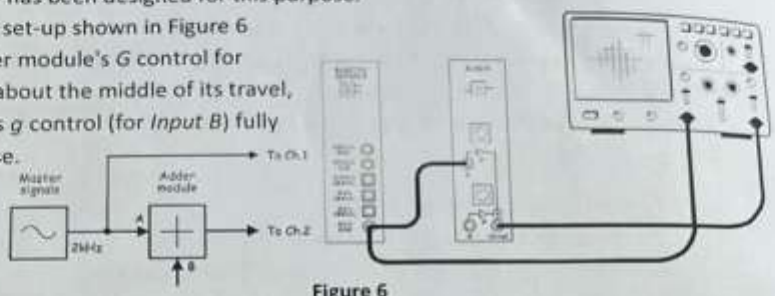


Figure 6

c. Adjust the scope's Time base control to view two cycles of the Master Signals module's 2kHz SINE output.

d. Set the scope's Mode control to the DUAL position to view the Adder module's output as well as the Master Signals module's 2kHz SINE output.

e. Set both of the scope's Vertical Attenuation controls to the 1V/div position.

f. Vary the Adder module's G control left and right and observe the effect.

g. Measure the voltage on the Adder module's Input A. Record your measurement in Table.

h. Turn the Adder module's G control fully clockwise.

i. Measure and record the Adder module's output voltage. 7.92

j. Calculate and record the voltage gain of the Adder module's Input A. $G = \frac{7.92}{3.72} = 2.13$

k. Turn the Adder module's G control fully anti-clockwise.

l. Repeat steps i and j.

Note: input B and gain control (g) do similarly as input A and gain control (G).

Parameter		Input voltage	Output voltage	Gain
Input A	Maximum	3.72	7.92	2.13
	Minimum		80mV	0.021

80mV / 3.72

m. Now turn both of the Adder module's gain controls fully clockwise.

n. Connect the Master Signals module's *2kHz SINE* output to both of the Adder module's inputs.

o. Measure the Adder module's new output voltage.

Adder's output voltage = 15.2V

2. It's a useful skill in telecommunications to be able to measure the phase difference between signals. The following procedure can be used whenever you need to measure the phase difference between two signals. Practice this by getting you to measure the phase difference between the Master Signals module's 100kHz SINE and COSINE outputs.

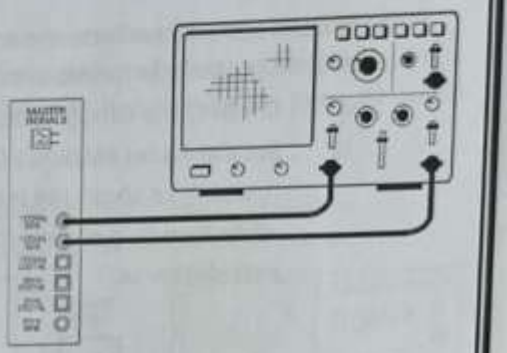


Figure 4

a. Connect the set-up shown in Figure 4.

b. Set the scope's Mode control to the Dual position to view both of the Master Signals module's 100kHz outputs.

c. Adjust the scope's *Time base* control to view as few **complete** cycles of the two signals as possible.

d. Set the *Input Coupling* controls for both channels to the *GND* position.

e. Use the *Vertical Position* control of both channels to align both traces with the horizontal line in the middle of the scope's display.

f. Set the *Input Coupling* control for Channel 1 to the *AC* position.

g. As accurately as you can, count the number of divisions for one cycle of the signal. Call this dimension the "period".
Record your measurement: period = ..2..

h. Set the *Input Coupling* control for Channel 2 to the *AC* position.

i. As accurately as you can, count the number of divisions between the two signals as shown in Figure 5. Call this dimension the "difference" and record your measurement here: difference = ...0.5..

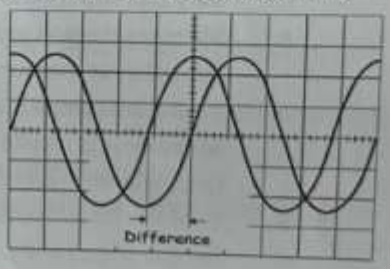


Figure 5

j. Use the equation $\theta_{difference} = \frac{difference}{period} \times 360°$, to calculate the phase difference between the two signals.
phase difference = ... $\frac{\frac{1}{2}}{2} \times 360 = 90°$

4. Several communications systems require that the signal to be transmitted (speech, music and/or video) is phase shifted. Crucial to being able to implement these systems in later experiments is the ability to phase shift any signal by almost any amount. The Phase Shifter module has been designed for this purpose.

a. Locate the Phase Shifter module and set its *Phase Change* control to the 0° position.
b. Set the Phase Shifter module's *Phase Adjust* control to about the middle of its travel.
c. Connect the set-up shown in Figure 7.

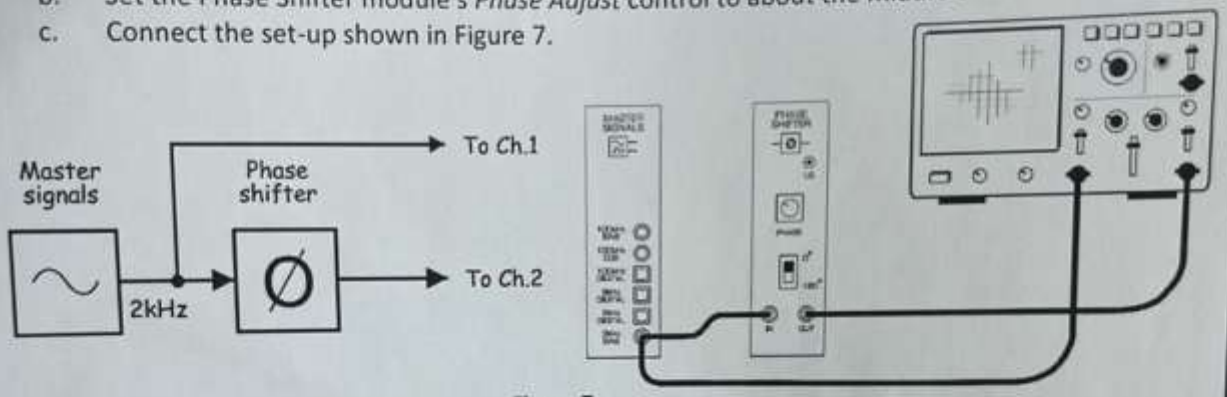


Figure 7

d. Adjust the scope's Time base control to view two cycles of the Master Signals module's 2kHz SINE output.
e. Set the scope's Mode control to the DUAL position to view the Phase Shifter's output as well as the Master Signals module's output.
f. Vary the Phase Shifter module's Phase Adjust control left and right and observe the effect on the two signals.
g. Set the Phase Shifter module's Phase Change control to the 180° position.
h. Vary the Phase Shifter module's Phase Adjust control left and right and observe the effect on the two signals then answer the following questions:

1- Does the module's output signal lag or lead the input signal, or it can do both? both

2- Is the phase shifter module capable of shifting a signal by 360°? Why? you can shift 180° directly & then move the shift knob to about another 180° so you can obtain a shift close to 360°

Pictures of output waveforms on oscilloscope







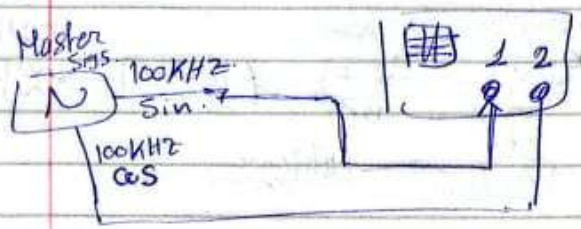




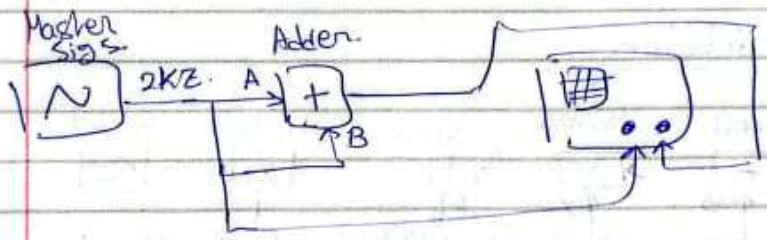
3.

ADC report 1
Block diagrams

1. Signal generation.



2. Signal addition.



3. Phase Shifting.

